Oxford Cambridge and RSA – Practice Paper

GCE Economics

H060/01 Microeconomics – AS Level

Practice Paper 3 (Hard) – Mark Scheme

This mark scheme follows the structure and marking principles of OCR H060/01 June 2024.

MARKING INSTRUCTIONS (summary)
1. Mark strictly to the mark scheme. Award marks that fairly reflect knowledge and skills demonstrated.
2. Use the level-of-response grids for parts (e)* and Section C. Start at the highest level and work down.
3. For Section C, if a candidate writes two answers, mark both and award the highest mark.
4. Award No Response (NR) if nothing is written; award zero if there is text but it is not worthy of credit.
5. Use the standard annotations: ✓ correct, KU, APP, AN, EVAL, BOD, TV, NAQ.

SECTION A – Multiple Choice

Question	Answer	AO	Rationale
1	**B**	AO1	A A privately owned car is a pure private good (rivalrous and excludable). B Correct: a toll motorway is excludable (via the toll) and only partially rivalrous (rivalrous when congested) – the defining features of a quasi-public good. C National defence is a pure public good (non-excludable, non-rivalrous). D A restaurant meal is a pure private good.
2	**B**	AO1	A This is the condition for productive efficiency, not allocative. B Correct: allocative efficiency is achieved when P = MC, i.e. the price consumers are willing to pay equals the marginal cost of production. C Profit maximisation occurs where MR = MC, which only coincides with P = MC in perfect competition. D Total revenue exceeding total cost is the condition for supernormal profit, not allocative efficiency.
3	**C**	AO2	A A natural disaster destroying capital reduces productive capacity → PPF shifts inwards. B A fall in the labour force reduces resources → PPF shifts inwards. C Correct: a technological improvement raises productivity and would shift the PPF **outwards**, not inwards. This is therefore the option that would *not* cause an inward shift. D Long-term resource depletion reduces productive capacity → PPF shifts inwards.
4	**C**	AO1	A Beef and leather are in **joint supply**, not composite demand. B Cars and petrol are in **complementary demand**. C Correct: land has alternative uses (housing or farming) – this is the definition of composite demand. Using land for housing reduces the amount available for farming. D Tea and coffee are **substitutes**.
5	**A**	AO3 (QS)	A Correct: if both demand and supply rise but price falls, the rightward shift in supply must exceed the rightward shift in demand. Larger shift in S pushes price down. B This would push price upward. C Elasticities affect the size of price/quantity changes but cannot by themselves determine the direction of price change when both curves shift. D Equilibrium quantity must rise when both D and S increase.
6	**A**	AO3 (QS)	MSC = MPC + MEC = £10 + £4 = £14. MSB = £14. Therefore MSC = MSB at Q = 100. A Correct: at MSC = MSB the market is allocatively efficient. The note 'MSB = MPC + MEC' is equivalent to 'MSB = MSC' which is the optimality condition. B Incorrect: MSB does not exceed MSC here (they are equal). C Incorrect: MSB is not less than MSC. D Although MPB and MPC are not given separately, the question concerns social efficiency.
7	**A**	AO2 (QS)	%ΔQ = PED × %ΔP = –0.5 × 8% = –**4%**. With |PED| < 1 (inelastic), revenue rises when price rises. Check: Original revenue 1 × 100 = 100. New revenue 1.08 × 96 = 103.68 → revenue rises. A Correct: quantity falls by 4%; total revenue rises. B Wrong direction of revenue change. C/D Use elasticity of –2.0 not –0.5.
8	**A**	AO2 (QS)	%ΔQ = (188 – 200)/200 × 100 = –**6%**. %ΔI = (33000 – 30000)/30000 × 100 = **+10%**. YED = –6/10 = –**0.60**. Bread is an inferior good (negative YED). A Correct: YED = –0.60. B Decimal-point error: 0.12/2 not adjusted to percentages. C Wrong sign – ignores the fall in quantity. D Inverts numerator and denominator.
9	**C**	AO1	A Price comparison is the consumer using available information, not asymmetric information. B This describes cost-push price-setting, not asymmetric information. C Correct: this is the classic 'lemons' problem – one party (the seller) knows more about product quality than the other (the buyer). Asymmetric information may lead to market failure. D Different technologies between firms is competition, not asymmetric information between buyer and seller.

Question	Answer	AO	Rationale
10	**A**	AO2	A Correct: buffer stocks work best when both PED and PES are inelastic, because price volatility is greatest and stockpiling is feasible. Storable, non-perishable, inelastically supplied goods are ideal candidates. B If demand is highly elastic, a buffer stock is less needed because consumers switch to substitutes when prices rise. C High storage costs make the scheme uneconomic. D Perishable goods cannot be stored, so buffer-stock schemes fail.
11	**B**	AO1	A Consumer preferences expressed through prices describes a market economy. B Correct: in a planned/command economy, the government decides what, how and for whom to produce. C Profit maximisation by firms is a market-economy feature. D Supply and demand interaction is the price mechanism in a market economy.
12	**C**	AO3	A The relevant triangle for welfare loss uses MSC and MSB, not MPB and MPC. B The vertical gap between MPC and MSC *is* the marginal external cost, but the welfare-loss triangle is bounded by MSC and MSB (or MPB) between Q* and Qm, not by MPC and MSC. C Correct: with a negative production externality, the welfare-loss (deadweight loss) triangle lies between the MSC and MSB curves from the social optimum quantity Q* up to the free-market quantity Qm. D The vertical distance is the marginal external cost, not the welfare loss area.
13	**B**	AO2 (QS)	$\%\Delta Q = (5.6 – 5)/5 \times 100 =$ **+12%**. $\%\Delta P = (460 – 400)/400 \times 100 =$ **+15%**. PES = 12/15 = **0.80** (inelastic supply). A Inverts and mishandles the calculation. B Correct: PES = 0.80. C Inverts the ratio (15/12). D Multiplies rather than divides percentages.
14	**B**	AO1	A Specialisation generally *increases* economies of scale. B Correct: specialisation increases dependence on a narrow range of products; a shock affecting that industry (e.g. drought, collapse in world price) has disproportionately large effects on the economy. C Specialisation typically *raises* productivity through learning-by-doing. D Specialisation through comparative advantage *raises* overall world output.
15	**A**	AO1	A Correct: acceptability (people are willing to receive it), divisibility (into small units for transactions) and portability (easy to carry) are essential properties of money. Durability and limited supply are also relevant. B Intrinsic value is not required (fiat money has none); beauty is irrelevant. C Government issuance helps acceptability but is not strictly necessary (private currencies exist); gold convertibility is not required since the end of the gold standard. D High transport costs would be a *disadvantage* – portability is required.

SECTION B – Question 16

Question	Answer	Mark	Guidance
16 (a)	Using Fig. 1, state the percentage of total free sugar intake by UK children that comes from soft drinks. **27% (1)**	**1**	Accept 27 without the % sign. Annotate with ✓. **Quantitative skills rewarded.**
16 (b) (i)	Using Fig. 2, calculate the % decrease in UK sugar consumption from soft drinks between 2015 and 2021. (500 – 360) / 500 (1) × 100 = **28% (2)** Correct method (1)	**2**	Accept 28% or 28 alone for 2 marks. Correct method without ×100 = 1 mark. Accept values within ±2 percentage points. **QS rewarded.**
16 (b) (ii)	Using PED = –0.8 for sugary soft drinks, explain how the SDIL will affect total expenditure on these drinks. PED is inelastic (|PED| < 1) (1). The tax raises price; quantity falls proportionally less than the price rise, so total expenditure on sugary soft drinks will **rise** (1). *OR (full numerical):* A 10% rise in price reduces Q by only 8%, so spending rises (2).	**2**	Accept reference to expenditure being almost unchanged because the levy was largely absorbed by reformulation (per the passage: 'total expenditure remained broadly unchanged'). Annotate with ✓.
16 (c)	**Using a demand and supply diagram, explain how an indirect tax such as the SDIL affects the market for sugary soft drinks.** Up to 3 marks for diagram: • Accurate labelling of axes (1) • Accurate labelling of supply curves with correct vertical (parallel) leftward shift of supply S → S+tax, with the vertical distance equal to the tax (1) • Accurate labels of old and new equilibrium (P,Q) and (P1,Q1) (1) Up to 2 explanation marks: • Tax raises producers' costs of supply, shifting S left/up by the amount of the tax (1) • Equilibrium price rises (P → P1) and quantity falls (Q → Q1) (1) • Reference to tax incidence: consumers bear more burden when PED is inelastic (–0.8) (1) *Maximum 4 marks overall.*	**4**	Annotate with ✓. Diagram must show a parallel upward shift of supply with movement along (inelastic) demand. Accept either S+tax or S1 as the new supply curve label.

Question	Answer	Mark	Guidance
16 (d) (i)	Using XED between sugary and sugar-free drinks = +1.4, explain the relationship between these two goods. XED is positive → sugary drinks and sugar-free alternatives are **substitutes** / in competitive demand (1). Value of 1.4 (greater than 1) indicates the relationship is **strong**: a 10% rise in the price of sugary drinks would lead to a 14% rise in demand for sugar-free alternatives (1).	**2**	Accept any clear explanation that distinguishes weak (|XED|<1) from strong (|XED|>1) substitutes.
16 (d) (ii)	Explain how the consumption of sugary drinks may lead to market failure. • Consumption of sugary drinks generates external costs to third parties – higher NHS treatment costs for obesity, type 2 diabetes and tooth decay, paid by all taxpayers (1) • Marginal social benefit (MSB) is therefore less than marginal private benefit (MPB) – sugary drinks are a **demerit good** (1) • Information failure: consumers may underestimate the long-term health risks and over-consume (1) • Free-market consumption (where MPB = MPC) exceeds the social optimum (where MSB = MSC) → over-consumption, welfare loss, allocative inefficiency (1) *Alternatively, up to 3 marks for a correctly labelled MPB/MSB diagram showing the welfare loss triangle.*	**4**	Candidate must link external costs / information failure to over-consumption for full marks. Annotate with ✓.

16 (e)* – Levels of response (10 marks)

Level / mark	Descriptor
Level 3 (7–10 marks)	**Reasonable application**; **strong analysis** with well-developed chains of reasoning; accurate and integrated diagram(s); **strong evaluation** with logical, well-supported judgement.
Level 2 (4–6 marks)	**Reasonable application**; **reasonable analysis** with single links; diagram errors affecting validity; **reasonable evaluation**; unsupported judgement.
Level 1 (1–3 marks)	**Limited application**; **limited analysis** lacking structure; diagrams absent or with significant errors; **limited evaluation**; stated or absent judgement.
0 marks	Response is not worthy of credit.

16 (e)*: Evaluate, using an appropriate diagram(s), the effects of the Soft Drinks Industry Levy on UK consumers.

The side of the argument presented first should be credited as Analysis with the other side credited as Evaluation.

Application
- SDIL introduced April 2018; 18p/litre and 24p/litre rates depending on sugar content
- PED for sugary soft drinks = –0.8 (inelastic); YED = +0.2; XED with sugar-free drinks = +1.4 (strong substitutes)
- Sugar consumption from soft drinks fell ~28% between 2015 and 2020
- Total expenditure on soft drinks broadly unchanged (consumers trading up / substitution)
- NHS costs from diet-related disease exceed £6bn p.a.

Analysis – Negative effects on consumers
- Diagram: SDIL shifts supply left (S → S+tax) raising price (P → P1) and reducing Q (Q → Q1); shaded area = tax burden
- Because PED is inelastic (–0.8), consumers bear most of the incidence – the price paid rises by more than Q falls
- Consumer surplus falls (welfare loss to consumers); regressive impact because lower-income households spend a larger share of income on sugary drinks
- Loss of consumer sovereignty – consumers are pushed away from preferred products by government policy
- Substitution towards sugar-free alternatives (high XED = +1.4) means consumers may not have suffered a large welfare loss in aggregate

Evaluation – Positive effects on consumers
- Long-term health benefits: lower sugar intake reduces obesity, type 2 diabetes, tooth decay (28% fall in sugar consumption from soft drinks)
- Manufacturers reformulated rather than passing on the tax – about half of products were reformulated below the threshold, so consumers paid little extra
- Information / nudge effect: the levy raised awareness of sugar content, helping consumers make better-informed choices (corrects information failure)
- External benefits internalised: lower NHS costs reduce taxes on everyone
- Total expenditure unchanged means the overall *financial* impact on consumers was modest, even if regressively distributed
- PED = –0.8 means the tax does change behaviour (4–8% fall in quantity per 10% price rise), so the policy can be effective without being highly burdensome

Possible judgement(s)
- On balance, beneficial in the long term (health and welfare gains exceed short-run cost) but regressive in distribution

• Effectiveness depends on continued reformulation and whether consumers substitute towards sugar-free rather than other high-sugar products (e.g. confectionery)
• Alternative or complementary policies (advertising restrictions, information campaigns, education in schools) may achieve similar gains with less burden on consumers
• Short-run consumer welfare loss may be justified by long-run reductions in NHS spending and improved population health

SECTION C – Levels of response (20 marks)

Level / mark	Descriptor
Level 4 (16–20)	Good KU; strong application; strong analysis with well-developed chains and accurate integrated diagram(s); strong evaluation with logical and well-supported judgement.
Level 3 (11–15)	Good KU; good application; good analysis beyond single links; diagrams predominantly correct; good evaluation; unsupported judgement.
Level 2 (6–10)	Reasonable KU; reasonable application; reasonable analysis with single links; diagram errors affecting validity; reasonable evaluation; unsupported judgement.
Level 1 (1–5)	Limited KU; limited application; limited analysis lacking structure; diagrams absent or with significant errors; limited evaluation; stated or absent judgement.
0 marks	Response is not worthy of credit.

17*: Evaluate, using an appropriate diagram(s), the view that regulation is more effective than indirect taxation in correcting market failure in markets such as sugary drinks, alcohol and tobacco.

The side of the argument presented first should be credited as Analysis with the other side credited as Evaluation.

Knowledge and Understanding
- Demerit goods, negative consumption externalities and information failure
- Indirect taxes – specific (per unit) and ad valorem; tax incidence and PED/PES
- Regulation – minimum unit pricing, advertising bans, sales restrictions, age limits, labelling, reformulation requirements
- Social optimum and welfare loss; allocative efficiency

Application
- Soft Drinks Industry Levy (UK, 2018); tobacco duty; alcohol duty / minimum unit pricing (Scotland 2018)
- PED for sugary drinks = –0.8; cigarettes are highly inelastic; alcohol varies by category
- Advertising restrictions on tobacco; HFSS food advertising restrictions to children
- Plain packaging / health warnings on cigarettes and alcohol

Analysis – Case FOR regulation being more effective
- Regulation can directly limit quantity to the social optimum (e.g. sales ban below 18); a tax only indirectly reduces Q via the price mechanism
- For very inelastic goods (cigarettes), taxes raise revenue but have little effect on quantity; regulation can be more effective at changing behaviour
- Diagram: with negative consumption externality, regulation can shift consumption from Qm directly to Q* (social optimum), eliminating the welfare loss triangle
- Regulation addresses information failure directly (e.g. labelling, health warnings); tax does not inform consumers
- Regulation does not have the regressive distributional impact of indirect taxes
- Cross-elasticity (e.g. XED for sugary vs sugar-free = +1.4) means reformulation requirements force producers to change products, bypassing the price mechanism

Evaluation – Case AGAINST regulation / FOR taxation
- Indirect taxes raise government revenue which can be hypothecated to fund treatment / education (regulation costs the government without offsetting revenue)
- Taxes internalise the externality and use market prices as an ongoing incentive to reduce production / consumption / reformulate – evidence: ~50% of UK soft drinks reformulated post-SDIL
- Regulation can be evaded (black markets, smuggling, illegal sales) – particularly for tobacco and alcohol
- Regulation requires monitoring and enforcement – costly and prone to government failure

• Outright bans / quantity restrictions remove consumer choice; taxes preserve choice while raising the relative price
• The SDIL evidence (28% fall in sugar consumption with broadly unchanged expenditure) suggests taxation can be very effective when paired with substitution opportunities

Possible judgement(s)
• Most effective approach is a **combination** – tax to alter price signals, regulation for the most damaging activities, with information provision to reduce information failure
• Relative effectiveness depends on PED – very inelastic goods (cigarettes) respond little to tax, so regulation has a larger role
• Distribution matters: regulation tends to be less regressive than indirect tax
• Government failure risk: setting tax at the right level or banning the right activity both require good information about external costs
• Effectiveness varies by product: regulation may dominate for tobacco; tax may dominate for sugary drinks where reformulation is feasible

18*: Evaluate, using an appropriate diagram(s), the extent to which government intervention is effective in correcting market failure caused by negative externalities.

The side of the argument presented first should be credited as Analysis with the other side credited as Evaluation.

Knowledge and Understanding
• Negative production / consumption externalities; divergence between private and social cost / benefit
• Government intervention: indirect taxes (Pigouvian), tradable permits, regulation, subsidies for substitutes, information provision, extending property rights
• Government failure: imperfect information, administrative cost, unintended consequences, regulatory capture, political constraints
• Social optimum (MSC = MSB) and welfare loss

Application
• SDIL on sugary drinks (28% fall in sugar from soft drinks, but regressive)
• Carbon taxes / emissions trading scheme (EU ETS, UK ETS)
• Congestion charges (London, ULEZ)
• Tobacco / alcohol duty; advertising bans
• Subsidies for renewable energy, electric vehicles

Analysis – Intervention can be effective
• Diagram: with a negative production externality, an indirect tax equal to the marginal external cost shifts supply from MPC to MSC; output falls from Qm to Q*; welfare loss triangle is eliminated
• Internalises externality, restoring allocative efficiency (P = MSC = MSB)
• Raises revenue which can fund mitigation, treatment or education
• Provides continuing market-based incentive for innovation (e.g. cleaner production, reformulation)
• Regulation can directly cap quantity (e.g. emissions limits, smoking bans in public)
• Tradable permits achieve a chosen quantity at lowest cost, by letting the market allocate abatement
• Evidence: SDIL reformulation; sharp falls in smoking prevalence following tax + advertising bans

Evaluation – Risks of government failure
• Difficulty valuing the external cost accurately – tax may be set too high or too low (e.g. uncertainty over social cost of carbon)
• Imperfect information about market response (PED, supply-side reaction) – the SDIL was largely absorbed by reformulation, so price signals to consumers were weak
• Regressive impact of indirect taxes – disproportionate burden on low-income households
• Administrative and enforcement costs may exceed welfare gain
• Tax evasion, smuggling, black markets (cigarettes, fuel)
• Unintended consequences (e.g. substitution to alternative harmful products; production shifting abroad / 'carbon leakage')
• Regulatory capture and lobbying may dilute or distort the intervention
• Political constraints – difficult to set taxes at theoretically optimal levels
• Coasian/private solutions (property rights, bargaining) may sometimes outperform government action

Possible judgement(s)
• Effectiveness depends on accuracy of valuation, PED and PES, and ability to enforce
• A combination of instruments is usually most effective: tax + information + regulation
• Government intervention is generally *preferable to inaction* when external costs are large – but imperfect intervention may still beat no intervention, even if it does not reach the social optimum
• Costs of government failure must be weighed against costs of market failure – intervention should pass a cost–benefit test
• Effectiveness varies by market – best for measurable, sizable externalities (carbon, congestion); harder for diffuse / informational failures

ASSESSMENT OBJECTIVES GRID

Question	AO1	AO2	AO3	AO4	TOTAL	(QS)
1–15	7	6	2		**15**	(5)
16(a)	1(1)				**1**	(1)
16(b)(i)	1(1)	1(1)			**2**	(2)
16(b)(ii)	1	1			**2**	
16(c)	2	2(2)			**4**	(2)
16(d)(i)	1	1			**2**	
16(d)(ii)	2	2			**4**	
16(e)*		1	4	5	**10**	
17*/18*	3	4(2)	6(3)	7(3)	**20**	(8)
TOTAL	**18**	**18**	**12**	**12**	**60**	**18**